Supporting Information

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SI Results

As an additional measure of bias in our calls of essential genes, we assayed these calls for polar effects, in which transposon insertions disrupt expression of downstream genes. We reasoned that, if there were strong polar effects, mutations in a nonessential gene would be lethal if they stopped expression of the downstream essential gene in the transcript; thus, we would not expect to find an operon with a nonessential gene upstream of an essential gene. To check for this possibility, we examined all adjacent genes within annotated operons (45) for essentiality. There were 87 instances of nonessential genes upstream from essential genes and 125 instances of essential genes upstream of nonessential genes. The similar frequency of these two arrangements indicates that there are not pervasive polar effects of the transposon, and conclusions of essentiality within an operon can be made for individual genes.

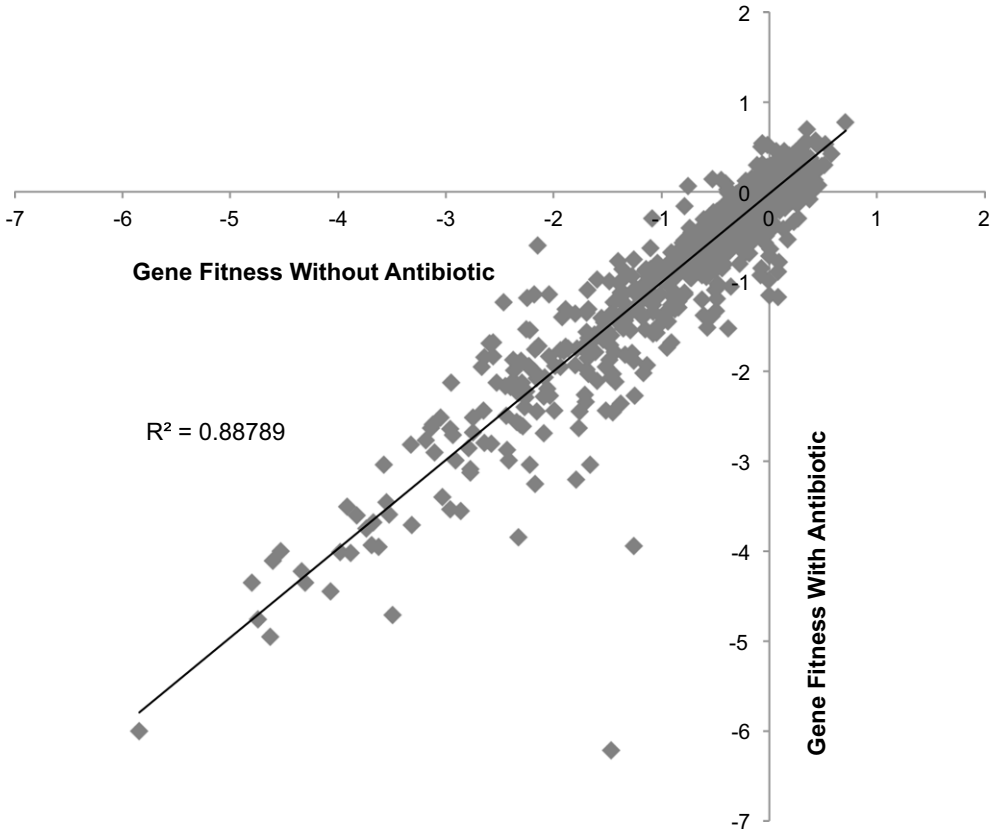


Fig. S1. Divergence of antibiotic and no antibiotic library outgrowths. Each gene's fitness score is plotted after outgrowth with kanamycin (*y* axis) or without kanamycin (*x* axis) for seven generations. Negative values represent genes that had negative effects when mutated (beneficial genes), and positive values represent genes that had positive effects when mutated (disadvantageous genes).

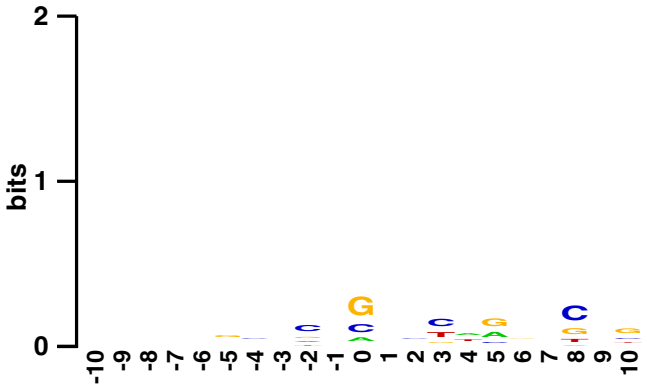


Fig. S2. Lack of sequence bias around insertion locations. The *y* axis shows bit score, a measure of sequence conservation, with a perfectly conserved base denoted by a score of two (24). The insertion location is between −1 and 0 bp.

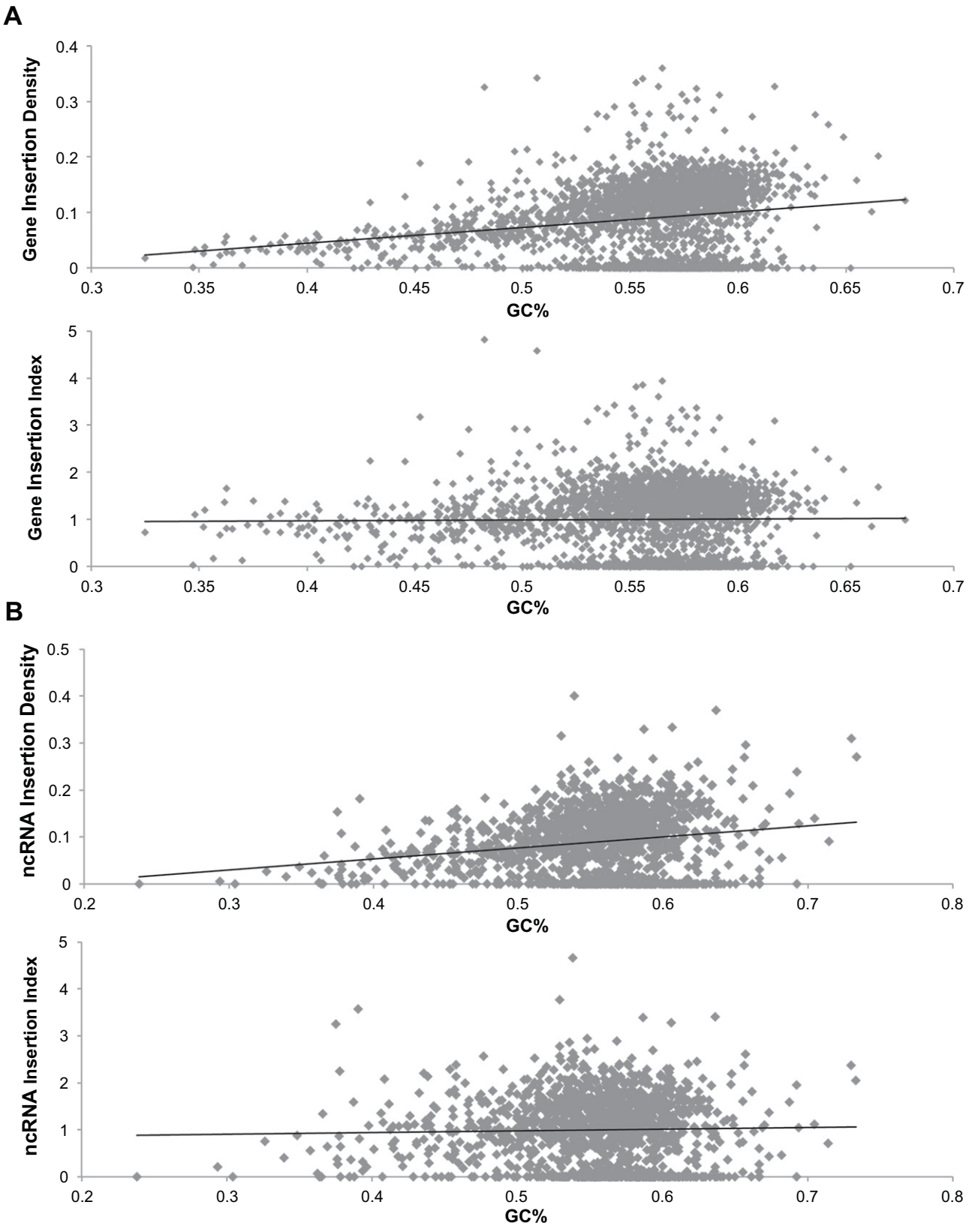


Fig. S3. Normalizing for GC bias of insertion density. *A*, *Upper* shows the insertion density (*y* axis) for each gene compared with its GC% (*x* axis) before normalization. These values were normalized by dividing each insertion density by the trend line value for its GC%. *A*, *Lower* shows the insertion density normalized for GC% (insertion index). *B*, *Upper* shows the insertion density for each of the recently annotated ncRNAs (45) compared with its GC% before normalization. *B*, *Lower* shows the insertion density normalized for GC% (insertion index).

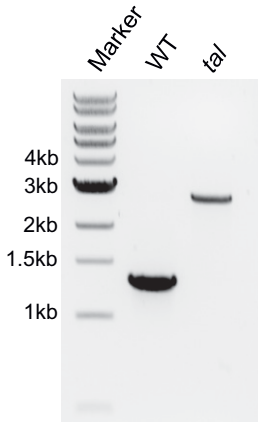


Fig. S4. Genotypic characterization of the recreated *tal* mutant. Lane 1, standard 1-kb ladder (New England BioLabs); lane 2, amplification of WT DNA with primers surrounding the *tal* gene; lane 3, amplification of *tal* mutant (UGS-3-C-11), which carries a 1.3-kb insertion, with the same primers. Each band is representative of three colonies tested.

Table S1. The major variations of techniques used for conjugation

Conjugation light intensity ($\mu mol \cdot m^{-2} \cdot s^{-1}$)	Ratio (mL:mL) of *Synechococcus elongatus*:*Escherichia coli*	Conjugation medium	Conjugation time (h)	Efficiency*
10–40	5:1	Filter (directly transferred to selective plate)	6–8	1.1×10^{-8} ($n = 1$)
10–40	2:1	Filter	6–8	1.5×10^{-8} ($n = 1$)
10–40	10:1	Filter	24	2.2×10^{-8} ($n = 1$)
10–40	10:1	Filter	6–8	3.3×10^{-8} ($n = 1$)
10–40	10:1	Agar transferred by wash to selective plate	24	4.4×10^{-8} ($n = 2$)
10–40	10:1	Agar transferred by wash to selective plate	6–8	4.7×10^{-8} ($n = 2$)
100–140	5:1	Filter	24	2.1×10^{-7} ($n = 1$)
100–140	5:1	Filter	6–8	2.1×10^{-7} ($n = 1$)
100–140	2:1	Filter	24	2.8×10^{-7} ($n = 1$)
10–40	1:1	Filter	6–8	3.0×10^{-7} ($n = 2$)
100–140	2:1	Filter	6–8	3.3×10^{-7} ($n = 1$)
10–40	1:1	Filter	24	5.0×10^{-7} ($n = 2$)
10–40	1:1	Agar with selective agent underlayed	24	7.8×10^{-7} ($n = 1$)
10–40	1:1	Filter	48	1.1×10^{-6} ($n = 1$)
100–140	1:1	Filter	24	2.0×10^{-6} ($n = 1$)
100–140	1:1	Filter	6–8	3.9×10^{-6} ($n = 4$)†

*Colony forming units (CFU) of conjugated mutants per CFUs of *S. elongatus* culture.
†The conditions used for generation of the RB-TnSeq library.

Table S2. Essential genes in central carbon metabolism

Synechococcus elongatus identification	Name	Pathway*	Essentiality	Conserved†	Redundant‡
SynPCC7942_2297	tal	Pentose phosphate pathway	Nonessential	1	0
SynPCC7942_0538	tktA	Pentose phosphate pathway	Essential	1	0
SynPCC7942_0604	cbbE	Pentose phosphate pathway	Essential	1	0
SynPCC7942_0584	rpiA	Pentose phosphate pathway	Essential	1	0
SynPCC7942_0529	pgl	Pentose phosphate pathway	Nonessential	1	0
SynPCC7942_2334	zwf	Pentose phosphate pathway	Nonessential	1	0
SynPCC7942_0039	gnd	Pentose phosphate pathway	Nonessential	1	0
SynPCC7942_0098	pyk	Glycolysis	Essential	1	0
SynPCC7942_0639	eno	Glycolysis	Essential	1	0
SynPCC7942_0469	pgmM	Glycolysis	Essential	1	1
SynPCC7942_1116	pgk	Glycolysis	Essential	1	0
SynPCC7942_1939	gap3	Glycolysis	Nonessential	0	1
SynPCC7942_0245	gap1	Glycolysis	Nonessential	0	1
SynPCC7942_1261	tpiA	Glycolysis	Essential	1	0
SynPCC7942_1443	cbbA	Glycolysis	Essential	1	0
SynPCC7942_0592	pfkA	Glycolysis	Nonessential	0	0
SynPCC7942_2029	pgi	Glycolysis	Essential	1	0
SynPCC7942_0781	ppsA	Glycolysis	Nonessential	0	0
SynPCC7942_2335	fbp	Glycolysis	Nonessential	0	1
SynPCC7942_0505	fbpI	Glycolysis	Essential	1	1
SynPCC7942_0485	pgam1	Glycolysis	Essential	1	1
SynPCC7942_1516	pgam2	Glycolysis	Essential	0	1
SynPCC7942_2078	pgam3	Glycolysis	Nonessential	0	1
SynPCC7942_0612	gltA	TCA cycle	Essential	1	0
SynPCC7942_0903	acnB	TCA cycle	Essential	0	0
SynPCC7942_1007	fumC	TCA cycle	Nonessential	1	0
SynPCC7942_1719	icd	TCA cycle	Essential	0	0
SynPCC7942_1533	sdhB	TCA cycle	Nonessential	0	0
SynPCC7942_1427	cbbS	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_1426	cbbL	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_1116	pgk	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_1742	gap2	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_1261	tpiA	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_1443	cbbA	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_2335	fbp	Calvin–Benson–Bassham cycle	Nonessential	0	1
SynPCC7942_0538	tktA	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_0505	fbpI	Calvin–Benson–Bassham cycle	Essential	1	1
SynPCC7942_0604	cbbE	Calvin–Benson–Bassham cycle	Essential	1	0
SynPCC7942_0977	prk	Calvin–Benson–Bassham cycle	Essential	0	0
SynPCC7942_0584	rpiA	Calvin–Benson–Bassham cycle	Essential	1	0

*Metabolic pathway as taken from the BioCyc database (31).
†Refers to conservation among all 13 of the cyanobacterial genomes studied in the work by Shi and Falkowski (26).
‡Functional redundancy as shown in the BioCyc database (31).

Table S3. Essential genes for the photosynthetic lifestyle

Synechococcus elongatus identification	Name	Description	Cyano conserved*
SynPCC7942_0027	bioF	8-Amino-7-oxononanoate synthase	No
SynPCC7942_0203	ribD	5-Amino-6-(5-phosphoribosylamino)uracil reductase…	Yes
SynPCC7942_0222		Hypothetical protein	No
SynPCC7942_0226		Sec-independent protein translocase TatA	No
SynPCC7942_0257		Protein of unknown function DUF92, transmembrane	Yes
SynPCC7942_0297	ftsH	FtsH peptidase homolog, chloroplast; metallopeptidase	No
SynPCC7942_0320	galE	UDP-galactose 4-epimerase	No
SynPCC7942_0322	ycf44	c-Type cytochrome biogenesis protein	Yes
SynPCC7942_0323	ccdA	Cytochrome *c* biogenesis protein-like	Yes
SynPCC7942_0330	atp1	Hypothetical protein	Yes
SynPCC7942_0333	atpG	F0F1 ATP synthase subunit B′	Yes
SynPCC7942_0439	chlM	Mg-protoporphyrin IX methyl transferase	Yes
SynPCC7942_0450	nifU	Putative NifU-like protein	Yes
SynPCC7942_0485	gpmB	Phosphoglycerate mutase	Yes
SynPCC7942_0492	ribF	Bifunctional riboflavin kinase/FMN adenylyltransferase	Yes
SynPCC7942_0537	fabF	3-Oxoacyl-(acyl carrier protein) synthase II	Yes
SynPCC7942_0560		ATPase	No
SynPCC7942_0584	rpiA	Ribose-5-phosphate isomerase A	Yes
SynPCC7942_0632	rplJ	50S ribosomal protein L10	Yes
SynPCC7942_0684	fabG	3-oxoacyl-[acyl-carrier-protein] reductase	Yes
SynPCC7942_0694	rps1	30S ribosomal protein S1	Yes
SynPCC7942_0775		Hypothetical protein	No
SynPCC7942_0853	dapL1	L,L-diaminopimelate aminotransferase	No
SynPCC7942_0884	EF-Tu	Elongation factor Tu	Yes
SynPCC7942_0894	aroK	Shikimate kinase	Yes
SynPCC7942_0897	minE	Cell division topological specificity factor MinE	Yes
SynPCC7942_0912	dnaX	DNA polymerase III, τ-subunit	Yes
SynPCC7942_0928		Outer envelope membrane protein	No
SynPCC7942_0942	ftsH	FtsH peptidase homolog, chloroplast, metallo peptidase	No
SynPCC7942_0959	obgE	GTPase ObgE	Yes
SynPCC7942_0977	prk	Phosphoribulokinase	No
SynPCC7942_0978	petH	Ferredoxin-NADP oxidoreductase	Yes
SynPCC7942_0988	ycf54	Conserved hypothetical protein YCF54	No
SynPCC7942_1002	psaD	Photosystem I reaction center subunit II	Yes
SynPCC7942_1068	pdhC	Branched-chain α-keto acid dehydrogenase subunit E2	Yes
SynPCC7942_1083		Probable glycosyltransferase	Yes
SynPCC7942_1086	hemE	Uroporphyrinogen decarboxylase	Yes
SynPCC7942_1198	phdD	Dihydrolipoamide dehydrogenase	Yes
SynPCC7942_1232	petC	Cytochrome b6-f complex iron-sulfur subunit	Yes
SynPCC7942_1274	ycf37	TPR repeat	Yes
SynPCC7942_1351		HAD-superfamily hydrolase subfamily IA, variant 3	No
SynPCC7942_1359		Coenzyme F420 hydrogenase	No
SynPCC7942_1427	rbcS	Ribulose 1,5-bisphosphate carboxylase small subunit	Yes
SynPCC7942_1457	plsC	1-Acyl-sn-glycerol-3-phosphate acyltransferase	Yes
SynPCC7942_1497		Hypothetical protein	No
SynPCC7942_1499	petF	Ferredoxin (2Fe-2S)	No
SynPCC7942_1520	rpsT	30S ribosomal protein S20	Yes
SynPCC7942_1590		Hypothetical protein	No
SynPCC7942_1673		Hypothetical protein	Yes
SynPCC7942_1734	ftrC	Ferredoxin-thioredoxin reductase catalytic chain	No
SynPCC7942_1737	sufD	Iron-regulated ABC transporter permease protein SufD	Yes
SynPCC7942_1830	trxA	Thioredoxin	Yes
SynPCC7942_1883	ycf53	Conserved hypothetical protein YCF53	No
SynPCC7942_1907	acsF	Mg-protoporphyrin IX monomethyl ester (oxidative) cyclase	No
SynPCC7942_1959	proS	Prolyl-tRNA synthetase	Yes
SynPCC7942_1983	pds	ζ-Carotene desaturase/three-step phytoene desaturase	No
SynPCC7942_2017		Hypothetical protein	Yes
SynPCC7942_2062	crtL	Lycopene cyclase (CrtL-type)	No
SynPCC7942_2084	chlG	Bacteriochlorophyll/chlorophyll a synthase	Yes
SynPCC7942_2089		Thioredoxin domain 2	Yes
SynPCC7942_2113	kprS	Ribose-phosphate pyrophosphokinase	Yes
SynPCC7942_2117	gatA	Aspartyl/glutamyl-tRNA amidotransferase subunit A	Yes

Table S3. Cont.

Synechococcus elongatus identification	Name	Description	Cyano conserved*
SynPCC7942_2136	dapB	Dihydrodipicolinate reductase	Yes
SynPCC7942_2274	chlD	Protoporphyrin IX magnesium-chelatase	Yes
SynPCC7942_2312	murG	UDP-*N*-acetylglucosamine-*N*-acetylmuramyl-(pentapeptide)...	Yes
SynPCC7942_2330	ctpA	C-terminal processing peptidase-2; serine peptidase	No
SynPCC7942_2359	nha3	Na^+/H+ antiporter	No
SynPCC7942_2415	lysS	Lysyl-tRNA synthetase	Yes
SynPCC7942_2503	por	Protochlorophyllide oxidoreductase	Yes
SynPCC7942_2524	tig	Trigger factor	Yes
SynPCC7942_2531	tsf	Elongation factor Ts	Yes
SynPCC7942_2537	clpPIII	ATP-dependent Clp protease proteolytic subunit	Yes
SynPCC7942_2538	clpR	ATP-dependent Clp protease-like protein	Yes
SynPCC7942_2581	petF	Ferredoxin (2Fe-2S)	Yes

S. elongatus essential genes conserved in the green lineage (plants and green algae) but not in nonphotosynthetic organisms as determined using the greencut2 dataset (30). GTPase, guanosine triphosphate hydrolase.
*Refers to conservation among all 13 of the cyanobacterial genomes studied in the work by Shi and Falkowski (26).

Other Supporting Information Files

Dataset S1 (XLSX)
Dataset S2 (XLSX)
Dataset S3 (XLSX)
Dataset S4 (XLSX)
Dataset S5 (XLSX)
Dataset S6 (XLSX)